# A Blockchain-Enabled Framework for Misinformation Monitoring

*Multilingual AI triage and a shared, tamper-evident ledger for South Africa's election fact-checkers*

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## 1. Introduction

In South Africa's 2024 election, AI deepfakes spread through WhatsApp groups and fact-checks were dismissed as propaganda [1]. Fact-checkers keep separate records, and detection tools rarely cover African languages:

**67%** of detection studies use English data [2]

**<10%** target low-resource languages [2]

**RESEARCH QUESTION**

*How can an integrated AI and blockchain framework enable tamper-resistant, transparent, multilingual monitoring and verification of election misinformation in South Africa?*

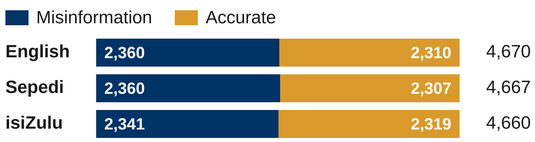
## 2. Methodology

### 2.1 A four-stage pipeline

1. **Ingest:** an Apache Kafka stream absorbs election-day surges.
2. **Triage:** AfroXLMR-large-76L [3] with LoRA [4] scores each claim.
3. **Verify:** every other organisation votes; ⌊E/2⌋ + 1 matching votes finalise, otherwise DISPUTED.
4. **Record:** Hyperledger Fabric [5] stores a SHA-256 hash; claim text stays in IPFS.

### 2.2 A synthetic multilingual corpus

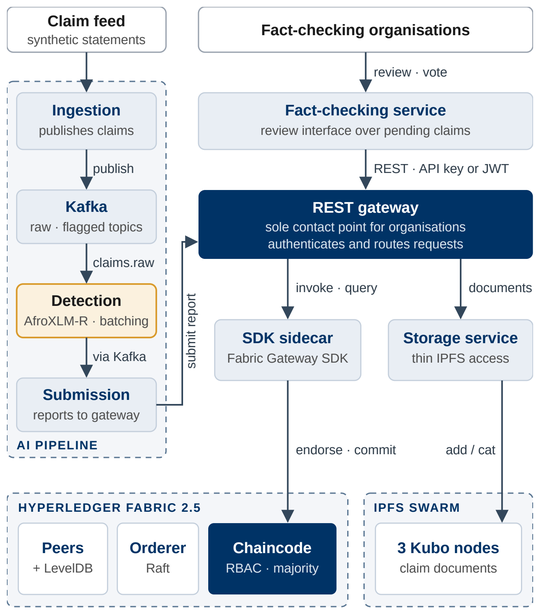
No Sepedi or isiZulu election corpus existed, so Gemini 3.6 Flash wrote paired true and false statements, machine-translated from English.




13,997 generated, 13,877 kept after cleaning; split 80/10/10.

## 3. System architecture

A permissioned **Hyperledger Fabric** consortium [5] with Raft ordering: no fees, deterministic finality, and roles enforced inside chaincode from X.509 certificate attributes.




Claims flow through Kafka and the AfroXLM-R detection service to the REST gateway, the organisations' only point of contact. Votes go on-chain; claim text stays in IPFS.

### 3.1 Roles enforced in chaincode

| Official | Fact-checker | Observer |
|---|---|---|
| submits, expires, admits members | submits reports and fact-checks | read-only audit |

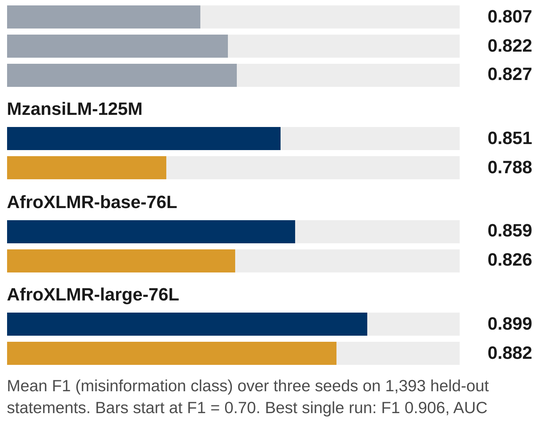
### 3.2 POPIA by design

- **s 10** Only hash, CID, AI metadata and org ID on-chain.
- **s 13** Fixed schema; no free-form personal fields.
- **s 14** Claim text off-chain in IPFS, so deletable.
- **s 19** SHA-256, TLS channels and chaincode RBAC.

## 4. Results

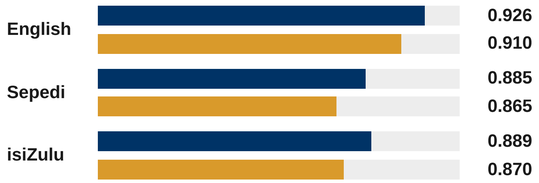



### 4.1 Misinformation detection, test F1




Mean F1 (misinformation class) over three seeds on 1,393 held-out statements. Bars start at F1 = 0.70. Best single run: F1 0.906, AUC 0.963.

### 4.2 F1 by language, AfroXLMR-large-76L




Same axis as 4.1. Sepedi and isiZulu trail English by only about 0.04.

### 4.3 LoRA's gap to full fine-tuning

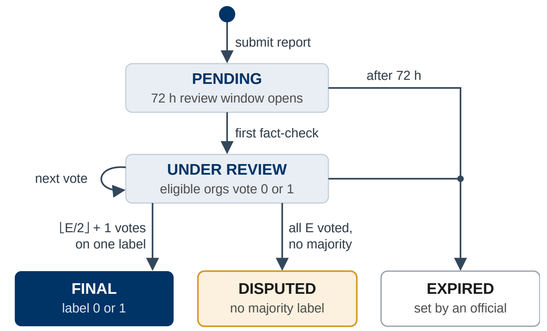



F1 lost by LoRA versus full fine-tuning (axis 0–0.07): the gap shrinks as models grow.

**WHY DEPLOY LORA?**

Two of three full fine-tuning runs overfitted (validation loss 0.40–0.49 above training loss); every LoRA run stayed within 0.04 while training only 1.29% of parameters.

## 5. Report lifecycle




E = organisations other than the submitter. The chaincode applies every transition; reports in the bottom row accept no further votes.

## 6. Conclusion

An Afrocentric encoder trained on translated synthetic data detects misinformation in Sepedi and isiZulu close to English; a Fabric consortium records joint verdicts tamper-evidently.

**Next steps:** real election posts, a multi-host deployment, a stakeholder web interface and per-user identities.

**Acknowledgements:** we thank our supervisors, **Dr. S Rananga** and **Mr. N Rananga**, and the DSFSI Research Group, University of Pretoria.

## 7. References

[1] G. Gondwe, "AI-driven disinformation and political influence on WhatsApp in South Africa's 2024 elections," *Int. J. Press/Politics*, 2025.

[2] S. Rananga, B. Isong, A. Modupe and V. Marivate, "Misinformation detection: a review for high and low-resource languages," *J. Inf. Syst. Informatics*, vol. 6, no. 4, 2024.

[3] J. O. Alabi, D. I. Adelani, M. Mosbach and D. Klakow, "Adapting pre-trained language models to African languages via multilingual adaptive fine-tuning," *COLING*, 2022.

[4] E. Hu et al., "LoRA: Low-rank adaptation of large language models," *ICLR*, 2022.

[5] E. Androulaki et al., "Hyperledger Fabric: a distributed operating system for permissioned blockchains," *EuroSys*, 2018.

**Code and data (QR code, top right):** github.com/SnowWoofer/A_Blockchain_Enabled_Framework_for_Misinformation_Monitoring